

STATEMENT OF NEED
460 CES
Cooling Water Repair Materials
Current as of 30 April 2026

1.0 REQUIREMENT TITLE: Pipe Repair and Fortification Materials

2.0 BACKGROUND

The 460th Civil Engineer Squadron (460 CES) is responsible for the maintenance and integrity in supporting the Mission Delta 4 (MD4) Restricted Area (RA) at Buckley Space Force Base. The cooling water system within this area is expected to experience leaks and requires immediate proactive fortification to ensure uninterrupted mission operations. This requirement is for a one-time, large-scale purchase of specialized materials necessary to execute these repairs and prevent future failures. These are pre-positioned spares.

3.0 SCOPE

The Government requires a contractor to provide three distinct material packages: an Elbow and Pipe Repair Package, a Pipe Freezing Kit, and a Carbon Fiber Composite Reinforcement Package. All materials must be new, commercial-off-the-shelf (COTS), and conform to the salient characteristics detailed below. The contractor shall be responsible for providing all items as a single, consolidated delivery.

4.0 SALIENT CHARACTERISTICS / SPECIFICATIONS

The delivered products shall meet or exceed the following minimum specifications:

4.1 ITEM 0001: Elbow and Pipe Repair Package (Quantity: 1 Package)

The package shall contain all of the following carbon steel pipe repair clamps:

- (2) 10-inch Straight Pipe Repair Clamps
- (2) 8-inch Straight Pipe Repair Clamps
- (2) 6-inch Straight Pipe Repair Clamps
- (2) 4-inch Straight Pipe Repair Clamps

- (2) 3-inch Straight Pipe Repair Clamps
- (2) 10-inch 90-degree Elbow Repair Clamps
- (2) 8-inch 90-degree Elbow Repair Clamps
- (2) 6-inch 90-degree Elbow Repair Clamps
- (2) 4-inch 90-degree Elbow Repair Clamps
- (2) 3-inch 90-degree Elbow Repair Clamps
- (2) 10-inch 45-degree Elbow Repair Clamps
- (2) 8-inch 45-degree Elbow Repair Clamps
- (2) 6-inch 45-degree Elbow Repair Clamps
- (2) 4-inch 45-degree Elbow Repair Clamps
- (2) 3-inch 45-degree Elbow Repair Clamps

Performance: All clamps shall be rated to withstand a minimum of 80 PSI water pressure.

Coating: All clamps shall be corrosion resistant, OR internally coated with an epoxy, OR be supplied with a compatible neoprene sheet for sealing.

Compliance: All materials in this package must be compliant with ASME PCC-2, ISO 24817, and/or ASME B13 standards for sealing high-pressure leaks.

4.2 ITEM 0002: Pipe Freezing Kit (Quantity: 1 Kit)

Function: Shall utilize either liquid carbon dioxide (CO₂) or liquid nitrogen (LN₂) to freeze stationary water, creating an in-line ice plug for pipe section isolation. Must be capable of forming dry ice at a temperature of -109°F or colder. This is a permanent tool for the RA maintenance shop.

Capability: Shall include all necessary jackets and components to support freezing pipes ranging from 3 inch to 10 inches in diameter.

Components: Shall include high-pressure hoses no less than 16 feet in length, all necessary CO₂ or LN₂ cylinder valve adapters, and associated timing boards and pads.

Compliance: The system shall be compliant with ASME PCC-2, ISO 24817, and/or ASME B13 standards for pipeline isolation.

4.3 ITEM 0003: Carbon Fiber Composite Reinforcement Package (Quantity: 1 Package)

System Type: Shall be a field-saturated composite system consisting of 150 ft of 4” to 12” width carbon fiber cloth, a compatible two-part epoxy, and a high-modulus filler material for sealing surfaces. The package must also include the necessary equipment for surface preparation to a near-white metal finish (SSPC-SP 10 or equivalent).

Application: Shall be designed to seal leaks in steel pipe in both hoop and axial directions on cooling water lines with diameters from 3” to 10”. The repair team will need to be prepared to repair live pipes.

Performance: Shall operate effectively at continuous temperatures up to 200°F.

Physical Properties: The cured ply thickness of the carbon fiber cloth shall be at least 0.0105 inches.

Compliance: The system shall be compliant with ASME PCC-2, ISO 24817, and/or ASME B13 standards for non-metallic composite repair systems.

5.0 DELIVERY AND ACCEPTANCE

5.1 Delivery Schedule: Due to the urgent nature of the required repairs, the contractor shall deliver all items no later than 14 days after receipt of order (ARO). Delivery time can be staggered for custom fabrications (e.g., 10" elbow clamps). The execution timeframe can also be amended to consider the driver screening window.

5.2 Acceptance: Items will be inspected upon delivery by authorized Government personnel to verify conformance with the salient characteristics listed in Section 4.0. The Government will be responsible for off-loading materials at the delivery location.

5.3 Delivery Location: Delivery to Buckley Space Force Base, Aurora. Driver must enter BSFB via Large Vehicle Inspection Point (LVIP) off 6th Avenue. Driver must meet standard 460 SFS base access criteria via a pre-vetting Entry Authority List (EAL); the government will provide. CES will support the contractor by submitting driver credentials to the Visitor Control Center (VCC) 2-weeks in advance of delivery. The contractor must provide a delivery driver authorized to enter BSFB legally.